



```
In [1]: import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
from sklearn.preprocessing import LabelEncoder
```

```
In [3]: data = pd.read_csv("loan_approval_dataset.csv")
print("First 5 rows:")
print(data.head())
print("\nColumn Names:")
print(data.columns)
print("\nDataset Shape:")
print(data.shape)
```

First 5 rows:

	loan_id	no_of_dependents	education	self_employed	income_annum	\
0	1	2	Graduate	No	9600000	
1	2	0	Not Graduate	Yes	4100000	
2	3	3	Graduate	No	9100000	
3	4	3	Graduate	No	8200000	
4	5	5	Not Graduate	Yes	9800000	

	loan_amount	loan_term	cibil_score	residential_assets_value	\
0	29900000	12	778	2400000	
1	12200000	8	417	2700000	
2	29700000	20	506	7100000	
3	30700000	8	467	18200000	
4	24200000	20	382	12400000	

	commercial_assets_value	luxury_assets_value	bank_asset_value	\
0	17600000	22700000	8000000	
1	2200000	8800000	3300000	
2	4500000	33300000	12800000	
3	3300000	23300000	7900000	
4	8200000	29400000	5000000	

	loan_status
0	Approved
1	Rejected
2	Rejected
3	Rejected
4	Rejected

Column Names:

```
Index(['loan_id', 'no_of_dependents', 'education', 'self_employed',
      'income_annum', 'loan_amount', 'loan_term', 'cibil_score',
      'residential_assets_value', 'commercial_assets_value',
      'luxury_assets_value', 'bank_asset_value', 'loan_status'],
      dtype='object')
```

Dataset Shape:

(4269, 13)

```
In [4]: le = LabelEncoder()
data[' education'] = le.fit_transform(data[' education'])
data[' self_employed'] = le.fit_transform(data[' self_employed'])
data[' loan_status'] = le.fit_transform(data[' loan_status'])
print("After Encoding:")
print(data.head())
```

After Encoding:

	loan_id	no_of_dependents	education	self_employed	income_annum \
0	1	2	0	0	9600000
1	2	0	1	1	4100000
2	3	3	0	0	9100000
3	4	3	0	0	8200000
4	5	5	1	1	9800000

	loan_amount	loan_term	cibil_score	residential_assets_value \
0	29900000	12	778	2400000
1	12200000	8	417	2700000
2	29700000	20	506	7100000
3	30700000	8	467	18200000
4	24200000	20	382	12400000

	commercial_assets_value	luxury_assets_value	bank_asset_value \
0	17600000	22700000	8000000
1	2200000	8800000	3300000
2	4500000	33300000	12800000
3	3300000	23300000	7900000
4	8200000	29400000	5000000

	loan_status
0	0
1	1
2	1
3	1
4	1

```
In [5]: data = data.drop('loan_id', axis=1)
print(data.columns)
```

```
Index(['no_of_dependents', 'education', 'self_employed', 'income_annum',
      'loan_amount', 'loan_term', 'cibil_score',
      'residential_assets_value', 'commercial_assets_value',
      'luxury_assets_value', 'bank_asset_value', 'loan_status'],
      dtype='object')
```

```
In [6]: X = data.drop('loan_status', axis=1)
Y = data['loan_status']
```

```
In [7]: X_train, X_test, Y_train, Y_test = train_test_split(
      X, Y, test_size=0.2, random_state=42
    )
```

```
In [8]: dt_gini = DecisionTreeClassifier(criterion='gini')
dt_gini.fit(X_train, Y_train)
```

```

gini_pred = dt_gini.predict(X_test)
print("Decision Tree (Gini) Predictions:")
print(gini_pred[:10])
gini_acc = accuracy_score(Y_test, gini_pred)
print("Decision Tree (Gini) Accuracy:", gini_acc)

```

Decision Tree (Gini) Predictions:
 [1 0 1 0 0 0 0 1 0 1]
 Decision Tree (Gini) Accuracy: 0.9765807962529274

```

In [9]: dt_entropy = DecisionTreeClassifier(criterion='entropy')
dt_entropy.fit(X_train, Y_train)
entropy_pred = dt_entropy.predict(X_test)
print("Decision Tree (Entropy) Predictions:")
print(entropy_pred[:10])
entropy_acc = accuracy_score(Y_test, entropy_pred)
print("Decision Tree (Entropy) Accuracy:", entropy_acc)

```

Decision Tree (Entropy) Predictions:
 [1 0 1 0 0 0 0 1 0 1]
 Decision Tree (Entropy) Accuracy: 0.9836065573770492

```

In [10]: rf_model = RandomForestClassifier(n_estimators=100)
rf_model.fit(X_train, Y_train)
rf_pred = rf_model.predict(X_test)
print("Random Forest Predictions:")
print(rf_pred[:10])
rf_accuracy = accuracy_score(Y_test, rf_pred)
print("Random Forest Accuracy:", rf_accuracy)

```

Random Forest Predictions:
 [1 0 1 0 0 0 0 1 0 1]
 Random Forest Accuracy: 0.9789227166276346

```

In [11]: print("\n--- Model Comparison ---")
print("Decision Tree (Gini):", gini_acc)
print("Decision Tree (Entropy):", entropy_acc)
print("Random Forest:", rf_accuracy)

```

--- Model Comparison ---
 Decision Tree (Gini): 0.9765807962529274
 Decision Tree (Entropy): 0.9836065573770492
 Random Forest: 0.9789227166276346

```

In [12]: new_data = [[2, 0, 0, 5000000, 15000000, 10, 699, 3000000, 2000000, 10000000],
new_data_df = pd.DataFrame(new_data, columns=X.columns)
prediction = rf_model.predict(new_data_df)
print("Prediction (0=Approved, 1=Rejected):", prediction[0])

```

Prediction (0=Approved, 1=Rejected): 0